



CS7.404: Digital Image Processing

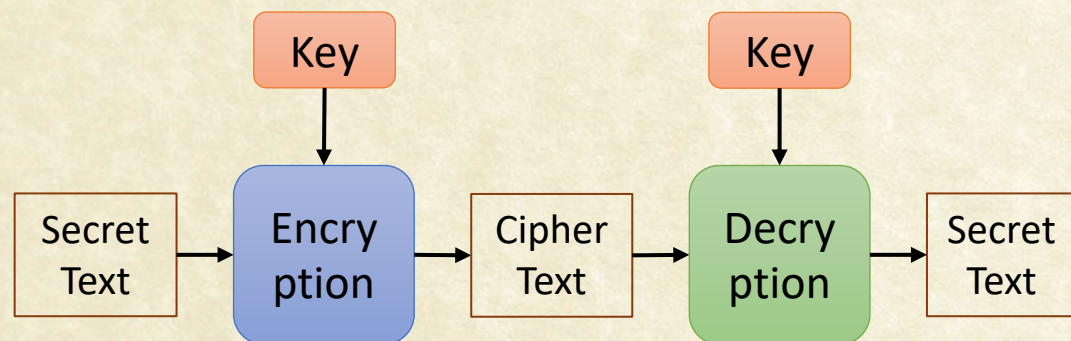
Monsoon 2025: **I**: Watermarking and Steganography

Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad

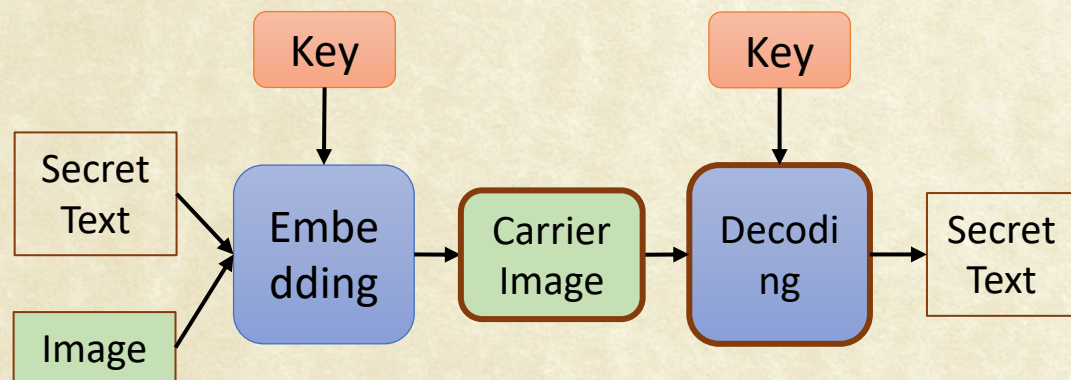


Related Fields

- Cryptography



- Steganography

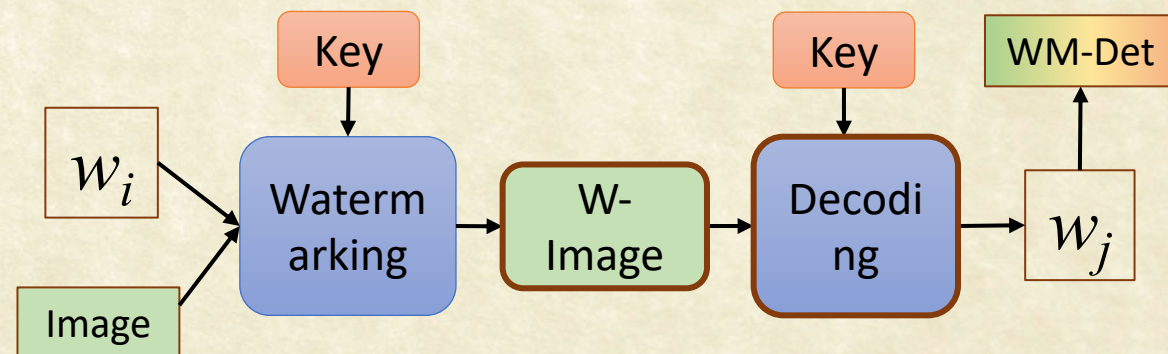


- Image Forensics

- Analyze images for deliberate or accidental watermarks
- Infer source, manipulations, etc.

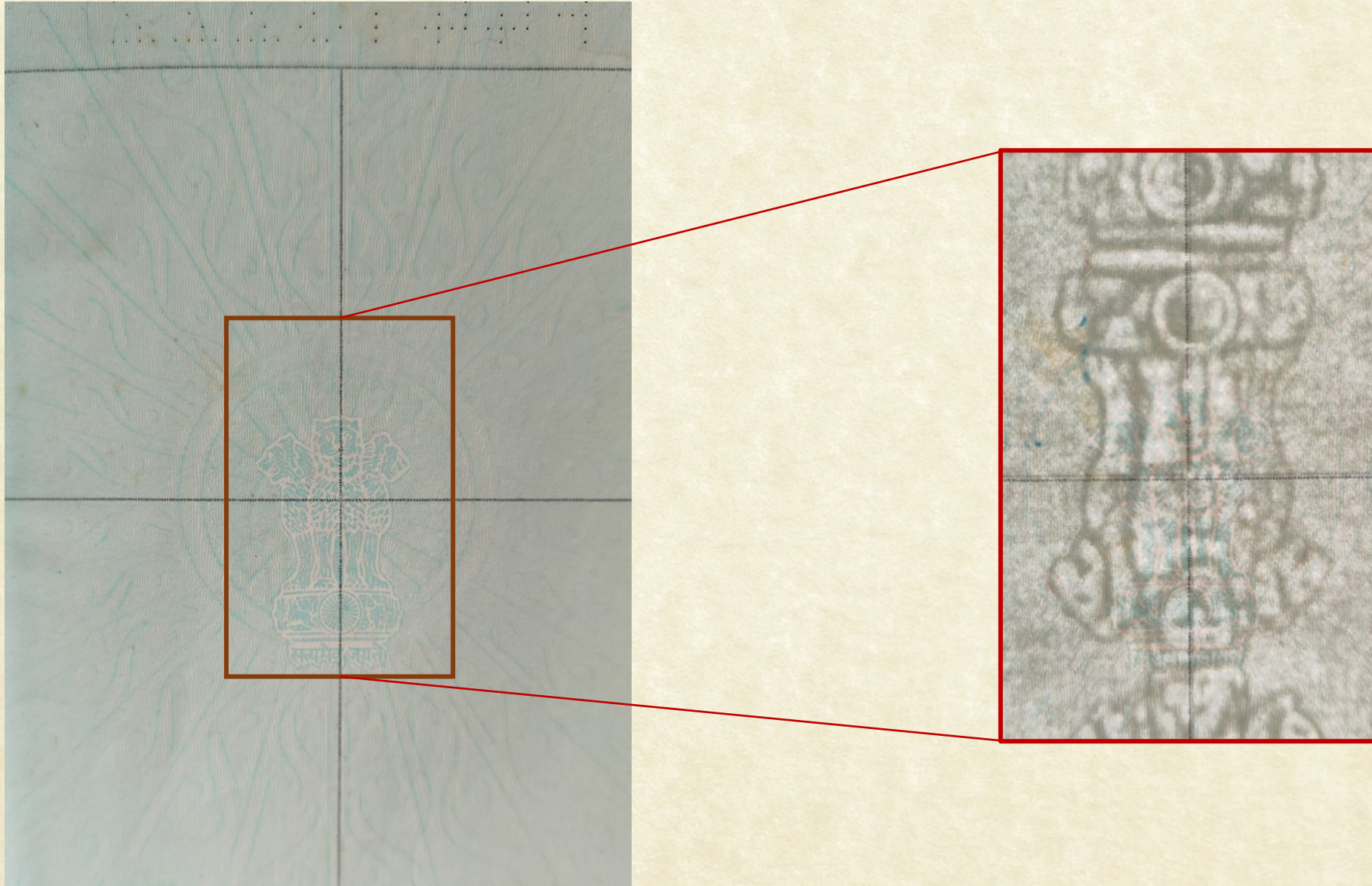
- Image Watermarking

- Goal: Protect or identify images



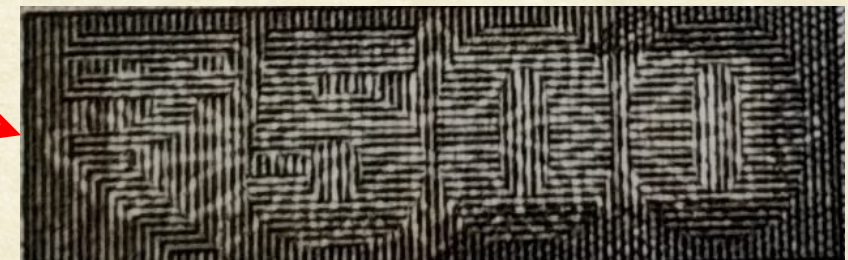
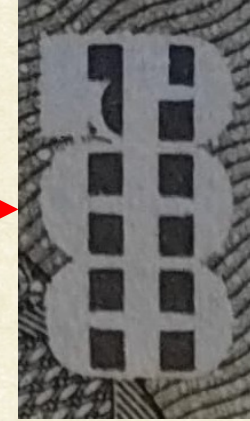


Watermarking: Physical





Watermarking: Currency



Goal: Establish authenticity
by making it difficult to copy



Digital Image Watermarking

- Copyright Identification
 - Watermark can be extracted even if the image is modified
 - Make the image less valuable if the watermark is removed
- Determine Authenticity
 - Watermark is destroyed even with small manipulation of image
- Preventing Copies
 - Patterns that can be detected by copying programs
- Tracing the Chain of Custody
 - Having different watermarks for each channel of distribution
 - Introduce watermarks on copying



Types of Watermarking

- Invisible vs. Visible
 - Invisible to human eye vs. Easily visible (masks some content)
- Robust vs. Fragile
 - Does not get destroyed even with significant image manipulations vs. Easily destroyed with even minor alterations
- Generic vs. Specific
 - Watermark is same across all images vs. Every image has a different watermark.





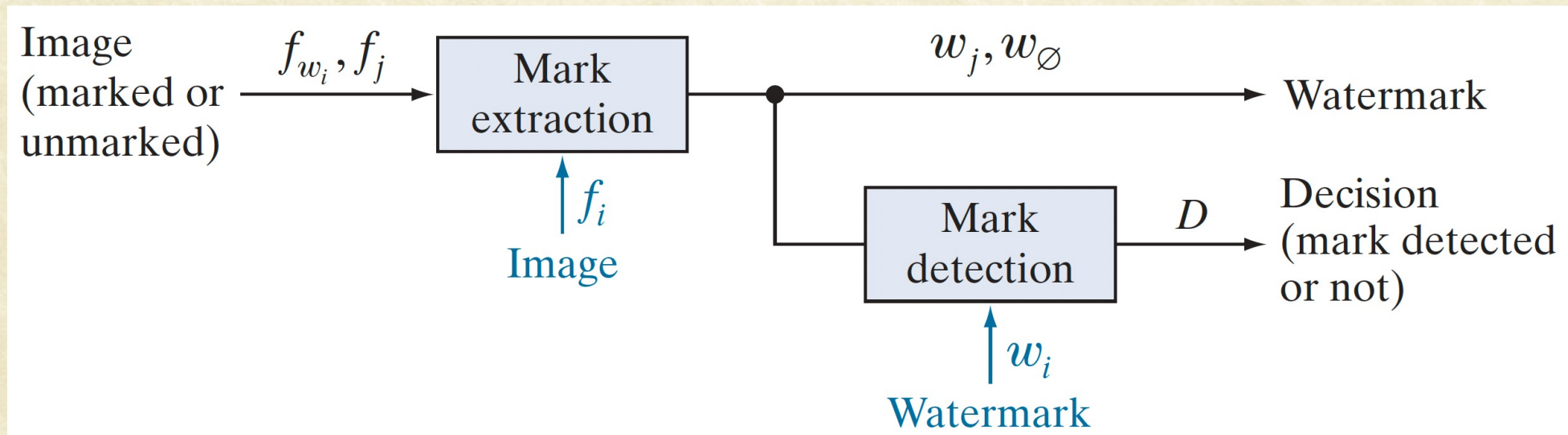
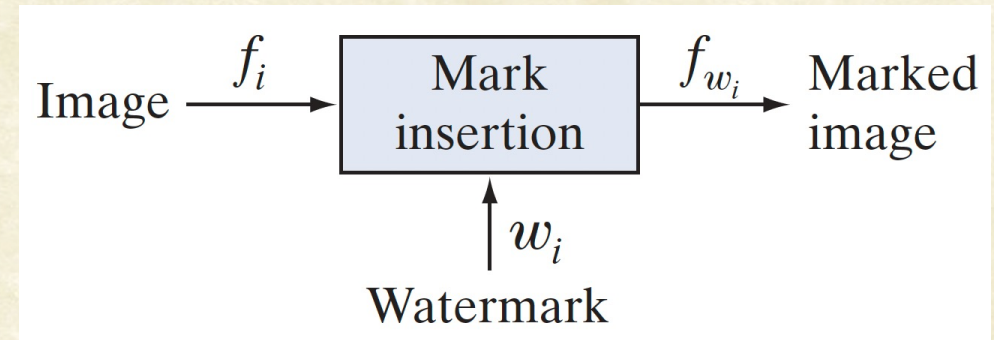
Desirable Properties of a Watermark

1. Visually Imperceptible
2. Statistically Similar
 - Makes it difficult to detect programmatically
3. Robust to Attacks (or Fragile)
 - Spatial transforms (scaling, rotation, cropping), Intensity transforms, Filtering, Adding noise, Print/Scan, Compression, Adding watermarks.
4. High Capacity
 - Number of watermark bits / pixel
5. Speed of Embedding / Detection



Watermarking Process

- Encoding phase inserts the watermark
- The decoding phase may or may-not use the original image and watermark.
 - Called *private/restricted-key* or *public/unrestricted-key*.





Visible Watermarking

- Superimpose a logo/mark on the image.

$$f_w(x, y) = (1 - \alpha)f(x, y) + \alpha \cdot w(x, y)$$

- $0 < \alpha < 1$ controls the visibility of the watermark
- α is set to 0 for pixels outside the watermark.



Perceivability

Statistical
Similarity

Robustness
to Attack

Capacity

Speed

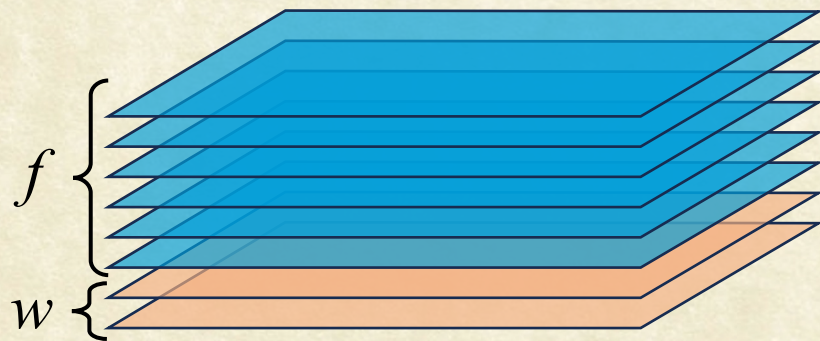


A Simple Invisible Watermarking

- Replace the least significant n (say 2) bits with the watermark.

$$f_w = 4 \left\lfloor \frac{f}{4} \right\rfloor + \frac{w}{64}$$

- Can be carried out using bit-wise operations



Perceivability
Statistical Similarity
Robustness to Attack
Capacity
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Robustness of LSB Watermarking

- LSB watermarking is not very robust, especially to operations like JPG compression

Image with watermark

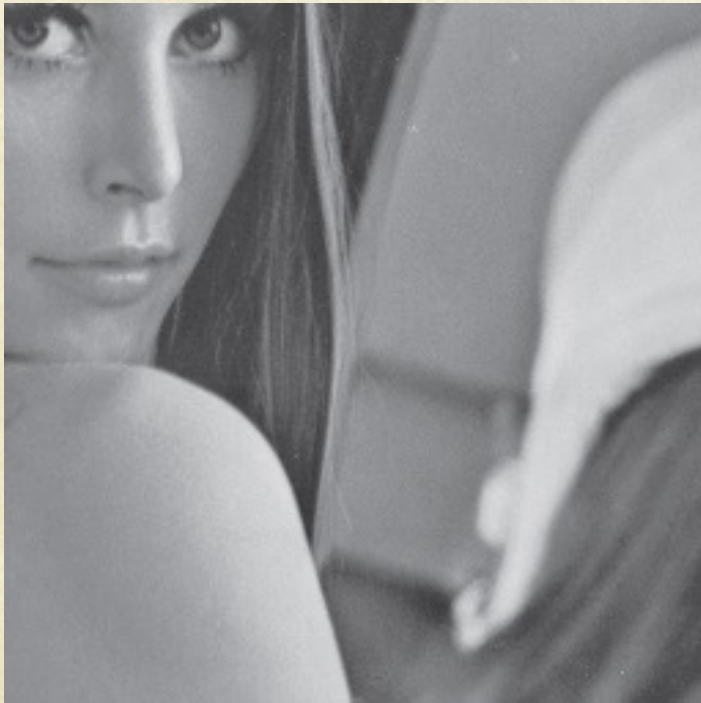
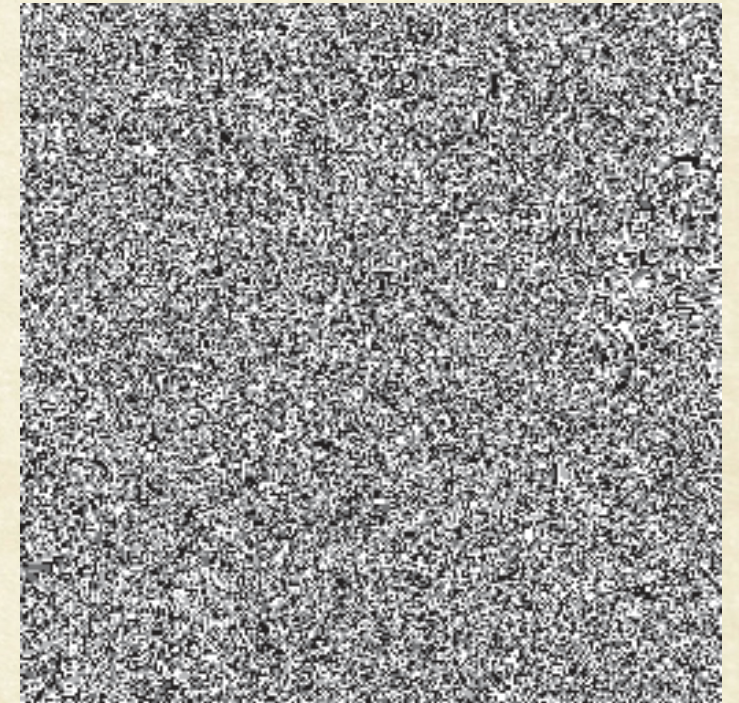


Image after JPEG compression



Recovered watermark





Adding Pseudo-Random Noise

- Divide the image into blocks of size $m \times n$.
- Generate two random noise patterns, W_0 and W_1 , both of size $m \times n$.
- To encode 0 or 1, add W_0 or W_1 to the chosen block.

$$B_W = B + \alpha W_i$$

- To decode; correlate each block with W_0 and W_1 , and the index with higher correlation is the decoded bit.
 - Note unmarked block will have similar correlation for both noise patterns.

Perceivability

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- | | | | | | | | | |
|----|----|----|----|-----|-----|-----|-----|------------|
| 16 | 11 | 10 | 16 | 24 | 40 | 51 | 61 | Capacity |
| 12 | 12 | 14 | 19 | 26 | 58 | 60 | 55 | |
| 14 | 13 | 16 | 24 | 40 | 57 | 69 | 56 | Efficiency |
| 14 | 17 | 22 | 29 | 51 | 87 | 80 | 62 | |
| 18 | 22 | 37 | 56 | 68 | 109 | 103 | 77 | Success |
| 24 | 35 | 55 | 64 | 81 | 104 | 113 | 92 | |
| 49 | 64 | 78 | 87 | 103 | 121 | 120 | 101 | |
| 72 | 92 | 95 | 98 | 112 | 100 | 103 | 99 | Speed |

16	11	10	16	24	40	51	61	Capacity
12	12	14	19	26	58	60	55	
14	13	16	24	40	57	69	56	Efficiency
14	17	22	29	51	87	80	62	
18	22	37	56	68	109	103	77	Success
24	35	55	64	81	104	113	92	
49	64	78	87	103	121	120	101	
72	92	95	98	112	100	103	99	Speed



A Robust DCT Watermarking (*Cox et al. [1997]*)

1. Compute the 2-D DCT of the image to be watermarked.
2. Locate its k largest (non-DC) coefficients, $c_1, c_2, \dots, c_k$, by magnitude.
3. Create a k -element pseudo-random sequence (watermark), $\omega_1, \omega_2, \dots, \omega_k$, taken from a Gaussian distribution $\sim N(0,1)$.
4. Embed the watermark from Step 3 by modifying the k largest DCT coefficients from Step 2 as:

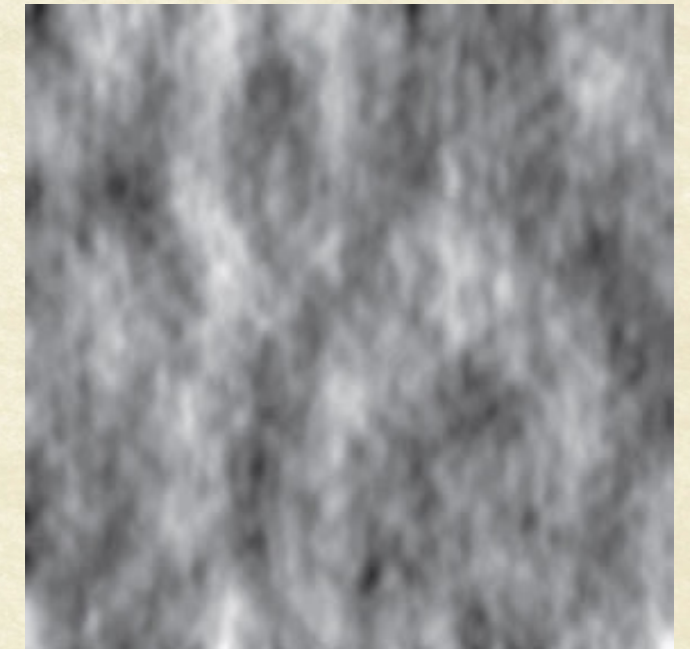
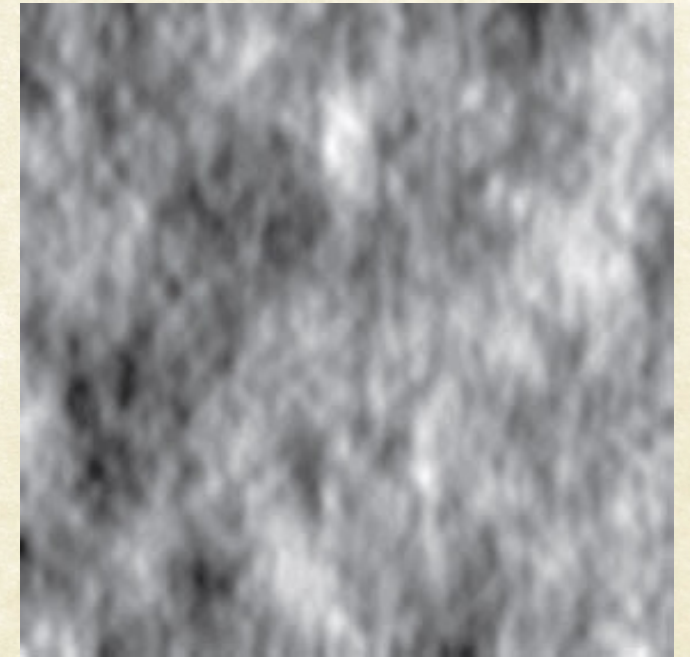
$$c'_i = c_i \cdot (1 + \alpha \omega_i), \quad 1 \leq i \leq k$$

- the constant $\alpha > 0$ controls the extent to which ω_i alters c_i .

5. Compute the inverse DCT of the result from Step 4.



Watermarking Examples



Two different watermarked images and their deviations from original (scaled). $\alpha = 0.1, k = 1000$.



Robustness and Perceivability

- By choosing the k largest coefficients, we can use small values for α , thus reducing perceived differences.
- As different DCT coefficients are modified, the effects are distributed both spatially and across frequencies.
- Attacks tend to degrade the image as they have to modify the more significant DCT components.

Perceivability
Statistical Similarity
Robustness to Attack
Capacity
Speed



Decoding and Detection

1. Compute the 2-D DCT of the image in question.
2. Extract the k DCT coefficients (in the positions corresponding to $c_1, c_2, \dots, c_k$, of Step 2 in watermarking:
 - denote them as $\hat{c}_1, \hat{c}_2, \dots, \hat{c}_k$.

3. Compute watermark, $\hat{\omega}_1, \hat{\omega}_2, \dots, \hat{\omega}_k$, using:

$$\hat{\omega}_i = \frac{\hat{c}_i - c_i}{\alpha c_i}, \quad 1 \leq i \leq k$$

$$\gamma = \frac{\sum_{i=1}^K (\hat{\omega}_i - \bar{\hat{\omega}})(\omega_i - \bar{\omega})}{\sqrt{\sum_{i=1}^K (\hat{\omega}_i - \bar{\hat{\omega}})^2 \cdot \sum_{i=1}^K (\omega_i - \bar{\omega})^2}}$$

4. Measure the similarity between $\hat{\omega}$ and ω using a metric such as correlation



Reference

- Ch 8, G&W textbook